

Gamifying Big Data Education: an RPG Approach with Integrated Quiz Assessments

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Abstract—The increasing need for Big Data education makes it harder to keep students interested and make sure they remember difficult ideas. By creating the educational role-playing game "Data Quest: Big Data Adventure," which incorporates quiz evaluations based on the Achievement-Based Unlocking System, this study tackles these problems. The goal of the study is to use an interactive learning platform to increase students' motivation and comprehension of big data. The study involved students from SMKN 10 Malang focus on 30 students selected specifically for pre-and post-tests to measure learning outcomes and motivation. Data was gathered through surveys, questionnaires, and pre- and post-tests. According to the statistical analysis, post-test scores significantly improved, gain score 0.54 overall across the five Big Data concepts (Volume, Variety, Velocity, Veracity, and Value). In terms of student interest, the results showed high levels of engagement with overall average score 81.19%, demonstrating the effectiveness of gamification as an engagement method. According to qualitative feedback, the game strengthened students' comprehension of difficult concepts while also being entertaining and instructive. According to the findings, incorporating quiz assessments into educational games through role-playing games offers a viable way to address student disengagement and enhance education learning outcomes.

Keywords—Gamification, Big Data Education, Role-Playing Game, Quiz Assessment

I. INTRODUCTION

Big Data's ascent has transformed a number of industries, including healthcare and business analytics, and increased demand for experts who can effectively evaluate and interpret massive datasets. Even though there is a growing demand for Big Data education, traditional teaching approaches frequently fall short in engaging students, making it challenging to communicate the intricate and abstract ideas that are central to the field. In order to improve learning experiences in Big Data education, educators are investigating alternative approaches like gamification [1].

The use of game elements in non-gaming contexts to increase students interests and learning outcomes are known as "gamification" in education [2]. According to research, educational games greatly increase students interests and learning outcomes by fostering a more engaging and entertaining learning environment [3]. Gamification offers a crucial dynamic to make learning more approachable and

interesting, especially in technical domains like Big Data where students may become overwhelmed by abstract concepts [4].

When using educational games, a number of factors affect how interest students are. These elements include the design of the game, its applicability to the curriculum, and its feedback system. According to studies, when educational games provide instant feedback on students' progress and are in line with their learning objectives, students are more likely to remain motivated [5]. Games can keep students' concentration while guaranteeing they retain important information by including quizzes and rewards [6].

Effective quiz assessments must be incorporated into educational games in order for them to be successful; gameplay components alone are not enough [7]. Instant feedback from well-designed tests enables students to assess their comprehension of difficult subjects like big data in real time. These tests reinforce important ideas and encourage active learning by acting as checkpoints during the learning process [8]. Quizzes should be balanced between testing students and making sure they retain important information without being overburdened in the context of Big Data education.

However, there are a number of issues with traditional Big Data teaching methods, including low student engagement, trouble remembering abstract ideas, and a dearth of interactive feedback. It's possible that lecture-based methods won't engage students enough to help them understand the subject matter [9]. These problems show how Big Data education needs to be more learner-centered and interactive.

Integrating quiz-based evaluations into an RPG educational game intended for Big Data learners is the aim of this study. This method offers students both engagement and a precise framework for assessing their comprehension by fusing structured assessments with narrative-driven gameplay. Throughout the game, the quizzes function as checkpoints that reinforce the concepts learned at each stage and provide real-time feedback [10].

This study offers a fresh take on Big Data education, which advances the field of educational technology more broadly. The goal of the research is to create a more effective and engaging approach to teaching complex subjects by

incorporating quiz assessments and RPG elements. Furthermore, the study may have an impact on the design of educational games in the future, especially in domains where mastery of intricate technical skills is necessary [11].

The first development of the RPG-based game, which focuses on incorporating quiz assessments designed for Big Data learning, marks the start of the research roadmap in 2024. In order to improve the gameplay and make sure the tests appropriately represent the curriculum's learning objectives, this phase will involve design iterations and user testing. The project will grow in the upcoming years in order to assess the game's scalability and potential for use in other technical education domains.

The absence of interactive, captivating tools in Big Data education that successfully integrate learning and assessment is the main issue this study aims to solve. The dynamic and responsive environment required to fully engage students in the subject matter is not provided by traditional methods [12]. By developing a gamified learning environment where tests incorporated into an RPG framework evaluate and reinforce students' comprehension of Big Data concepts in real time, this study seeks to address this problem.

II. LITERATURE REVIEW

A. Fixed Theoretical Basis

Constructivism and experiential learning theory are two important theoretical frameworks on which the idea of gamification in education is based [13]. According to constructivism, students actively create their own knowledge and understanding via interactions and experiences. Since gamification encourages active participation and interactive learning experiences that let students fully immerse themselves in the content, it fits in nicely with this theory [14]. Furthermore, experiential learning theory highlights the value of practical experiences in the learning process and contends that students learn best when they are able to apply their knowledge in real-world situations, as is the case in gamified classrooms.

B. Free Theoretical Basis

Numerous theories that support the use of game elements to increase motivation and engagement lend support to the implementation of gamification in education. Students are more engaged when they feel capable, independent, and connected to others, according to Self-Determination Theory (SDT), which highlights the importance of intrinsic motivation in learning [15]. Gamification meets these needs by introducing game mechanics like challenges and rewards, which may enhance learning results. Furthermore, the use of gamification in educational contexts is further supported by the Flow Theory, which explains the ideal state of engagement when an individual is totally absorbed in an activity [16].

C. Problem Review

One important issue that has received more concentration recently is the challenge of getting students interested in Big Data education. The need for experts in data analytics is only going to increase as Big Data becomes a more significant field in many different industries. However, low engagement and retention rates result from traditional teaching methods like lectures and textbooks, which frequently fail to pique students' interest. Investigating solutions like gamification with quiz assessments is essential because this problem

emphasizes the need for creative teaching strategies that can improve student interest and learning outcomes [17].

The low level of student participation in Big Data education has already been recognized by research as a serious issue. According to a number of studies, even though students may understand the value of Big Data skills, they find it difficult to stay motivated and involved when using conventional teaching methods. Their capacity to apply the knowledge in practical situations may suffer as a result of this disengagement, which may lead to a superficial grasp of the ideas [18].

By concentrating on two important areas, the issue of student disengagement in Big Data education can be resolved: by increasing the level of interaction in the learning process and offering ongoing evaluation to help students retain the material. We can create an organized strategy to improve comprehension and engagement by addressing these two issues. The perfect answer is provided by gamification, particularly through role-playing games (RPGs), which combine integrated quiz assessments with immersive learning experiences.

D. Theoretical Review

A gamified learning model has already been developed as part of the research methodology to address this problem. To create a dynamic and captivating learning environment, the model combines game mechanics like level progression and quizzes with instructional content. The Achievement-Based Unlocking System, which allows players to access quizzes as a form of evaluation only after finishing missions in each arc, is a crucial component of this model. This unlocking system encourages students to fully engage with the material before moving on by making sure that access to new quizzes or additional assessments is dependent upon completing the prior challenges [19].

In terms of data acquisition, the study uses in-game assessments to collect information on student performance. At different points during the game, these tests are intended to gauge students' comprehension of Big Data concepts and offer instant feedback. Analyzing quiz results to assess learning progress and pinpoint areas where students might struggle are two examples of data manipulation. After that, the findings are analyzed to modify the game's difficulty and content to better suit the needs of the learners [20].

E. Previous Research

Zainuddin (2020): Zainuddin [21] explored the role of gamified e-quizzes in formative assessment systems. Their research showed that integrating quizzes with game elements significantly enhanced student engagement and learning outcomes, indicating the effectiveness of gamification in promoting deeper understanding of complex subjects.

Nuci (2021): Nuci [22] examined the impact of game-based digital quizzes in online lectures, revealing that these quizzes improved student engagement during remote learning sessions. The study highlighted the importance of interactive assessments in maintaining student concentration and reinforcing learning concepts.

F. Research Hypothesis

Based on the literature and theoretical frameworks reviewed, the research hypothesis for this study is as follows:

Hypothesis 1: Gamification, particularly through RPGs with quiz assessments, can improve students' understanding compared to traditional methods.

Hypothesis 2: The integration of continuous assessment through quizzes within the gamified model will enhance both student engagement and retention of complex Big Data concepts.

Hypothesis 3: The methodology and results thus far indicate that this approach can increase student engagement and learning; however, extensive testing is required to fully validate its effectiveness across a larger student population.

III. METHODOLOGY

A. Research Design

The framework for addressing the research questions about how well gamification works to improve student engagement and comprehension in Big Data education is laid out in the study's research design. The design emphasizes the value of active engagement and practical applications in learning, and it is based on constructivist and experiential learning theories. The following are the main research questions:

- How does the integration of gamification through RPG elements impact students' understanding of Big Data concepts?
- What effects does the Quiz Assessments have on student interests and learning outcomes?

B. Participants

The study involved students from SMKN 10 Malang, with a focus on 30 students selected specifically for pre-and post-tests to measure learning outcomes and student interests. This subset of participants provided a focused view of the game's impact on understanding Big Data concepts, compared to traditional learning methods. As a group from a vocational school setting, these students offer valuable insights into the effectiveness of a gamified approach in contexts where practical application of Big Data knowledge is essential.

C. Data Collection and Analysis

To guarantee a comprehensive assessment of the educational impact of "Data Quest: Big Data Adventure," data collection will entail a comprehensive approach utilizing multiple methods.

Tests before and after: Pre- and post-tests will be used to gauge how well the game has improved students' comprehension of Big Data concepts. Before students play the game, the pre-test will evaluate their prior understanding of the five main Big Data concepts: volume, variety, velocity, veracity, and value. A post-test will be administered after they have finished the game to assess any shifts in their comprehension. Quantitative information demonstrating the gamification approach's educational impact will be provided by the test results.

Gain Score: Gain scores will be calculated to determine the degree of learning improvement, using the formula:

$$\text{Gain Score} = \frac{\text{Post-test Score} - \text{Pre-test Score}}{100 - \text{Pre-test Score}} \quad (1)$$

Gain scores will be categorized as follows to evaluate individual student progress: High Gain: ≥ 0.7 ; Moderate Gain: $0.3 \leq \text{gain} < 0.7$; Low Gain: < 0.3 . This categorization will provide insight into the effectiveness of the game-based learning approach compared to traditional methods.

Questionnaires and Surveys: To collect qualitative information on student interests, learning outcomes, general game fulfillment, surveys, and questionnaires will be employed in addition to tests. Likert scale questions will be incorporated into the surveys to enable students to indicate how much they agree or are satisfied with different aspects of the game, including its usability, degree of engagement, and perceived applicability of the content. In order to get thorough input regarding their experiences and any recommendations for enhancement, open-ended questions will also be incorporated.

Interpretation of Feedback: Thematic analysis will be used to examine the qualitative input obtained from open-ended survey questions. In order to gain insight into how students view the game and their educational experiences, this process entails locating recurring themes and patterns in the responses. The study attempts to obtain a thorough grasp of how the gamification strategy affects learning outcomes and student interests by combining quantitative and qualitative data.

D. Ethical Considerations

Conducting this research requires careful consideration of ethical issues. All participants and their guardians will be asked for their informed consent, guaranteeing that they are aware of the study's objectives and their legal rights. Through data anonymization and safe storage of all research materials, the study will place a high priority on participant confidentiality. In order to protect the study's integrity and the participants' welfare, the research will also adhere to ethical standards established by the educational institution and pertinent authorities.

IV. RESULTS

A. Development of "Data Quest: Big Data Adventure"

The educational role-playing game "Data Quest: Big Data Adventure" aims to teach students the fundamentals of big data, with a particular emphasis on the five V's: volume, variety, velocity, veracity, and value. Curating educational resources about these ideas and making sure they fit the target students' curriculum is the first step in the development process. Students can thoroughly examine each idea because each of the game's five arcs relates to one of the five Vs. Students will take part in tests created with the Achievement-Based Unlocking System after finishing each arc.



Fig. 1. Gameplay RPGs

The image above is an example of one of the Quiz Assessments of a game that is being developed.



Fig. 2. Evaluation Using Quiz Assessments

This system encourages mastery of the material before moving on to the next concept by limiting access to quizzes as evaluations to students who have successfully finished the corresponding arc.

B. Learning Outcomes

The pre-test and post-test were designed to evaluate the effectiveness of the gamified learning approach in "Data Quest: Big Data Adventure." The conditions under which these tests were conducted varied to reflect different teaching methodologies.

Pre-Test: The pre-test was administered following a traditional learning approach. Students were taught the material in a face-to-face classroom setting using conventional methods such as lectures and discussions. After the instruction, students completed a quiz to assess their understanding of the Big Data concepts.

Post-Test: The post-test, in contrast, was integrated into the gameplay experience of the educational game. Students participated in interactive learning through the game and completed evaluations in the form of quizzes embedded within the game. This approach allowed for a gamified assessment experience that aligned with the game's mechanics and storyline.

Students were given both pre-and post-tests to gauge the success of the gamified approach. The average scores of the players before and after playing the game are shown in the following table. Table I provides a summary of the findings.

TABLE I. PRE AND POST-TEST RESULT

Assessment Type	Average Score (Pre-Test)	Average Score (Post-Test)	Gain Score
Volume	55	78	0.51
Variety	60	82	0.55
Velocity	57	80	0.53
Veracity	59	83	0.58
Value	58	79	0.50
Overall	57.8	80.4	0.54

C. Gain Score Analysis

To evaluate the effectiveness of the gamified learning strategy implemented in "Data Quest: Big Data Adventure," the pre-test and post-test scores were analyzed using a gain score approach. This method helps measure the relative improvement in student performance, normalized by the initial scores to provide a clearer view of learning gains.

Descriptive Statistics: The pre-test and post-test results show notable improvement across all five V's of Big Data. The mean pre-test score was 57.8, while the mean post-test score increased to 80.4. These results correspond to an overall gain score of 0.54.

D. Student Interests

The evaluation of student interests in the context of Big Data education through the educational RPG 'Data Quest: Big Data Adventure' focused on five key dimensions: positive emotion, interest, focus, engagement, and learning completeness. The assessment aimed to measure the impact of gamification on student interests. The results of the evaluation are summarized in Table II below:

TABLE II. STUDENT INTERESTS RESULTS

Indicator	Result	Classification
Positive Emotion	82.50%	High
Interest	79.84%	Moderate
Focus	75.27%	Moderate
Engagement	83.13%	High
Learning Completeness	85.2%	High
Average	81.19%	High

Positive emotion (82.50%): the evaluation of positive emotion showed that students experienced a high level of emotional engagement with the educational RPG game. This indicates that the gamified approach used in 'Data Quest: Big Data Adventure' succeeded in generating a positive emotional response, contributing to an enjoyable learning experience. The high score reflects that students found the game both entertaining and motivating, which is a critical factor in maintaining student involvement and enthusiasm throughout the learning process [23].

Interest (79.84%): in terms of interest of students reported moderate levels of engagement with the game content. While this shows that the game captured the students' attention, there is an opportunity to further enhance the content's relevance or appeal to ensure higher sustained interest [24]. Students who scored lower in this category might benefit from additional content that more directly connects with their personal preferences or career aspirations within the Big Data field.

Focus (75.27%): the focus dimension, with students indicating moderate levels of attention, suggests that the game managed to engage most students but faced challenges in keeping all participants fully concentrated throughout the game [25]. This moderate result suggests that while the game was generally effective, certain gameplay elements could be adjusted to minimize distractions and improve sustained focus. Adding features such as short, interactive tasks or more dynamic feedback mechanisms could further enhance concentration.

Engagement (83.13%): engagement was particularly strong, with students reporting high levels of participation [26]. This result demonstrates that the game's interactive and rewarding elements were successful in maintaining students' involvement in the learning process. The Achievement-Based Unlocking System, where students progress through challenges by completing tasks, likely played a significant role in fostering this engagement by providing a clear sense of achievement and progression.

Learning completeness (85.2%): finally, learning completeness was high, with students demonstrating strong levels of mastery over the game content. This indicates that the game successfully supported the students in completing the educational tasks and mastering key Big Data concepts. The effectiveness of the gamification approach in promoting comprehensive learning outcomes reflects the potential for RPGs to provide both an enjoyable and educational experience.

E. Discussion

This study examined the effectiveness of the gamified educational RPG, "Data Quest: Big Data Adventure," in enhancing student understanding of Big Data concepts at SMKN 10 Malang, as measured through pre-and post-test assessments. The pre-test was conducted using traditional teaching methods, while the post-test was integrated into the game, reflecting the impact of gamification on students' learning outcomes.

In terms of student interest, the results showed high levels of engagement, particularly in learning completeness (85.2%) and engagement (83.13%), which directly contributed to higher gain scores. This finding indicates that students who were more engaged with the game tended to experience greater learning gains, reinforcing the idea that an engaging gamified environment can foster deeper learning.

Integrating a well-designed educational game in learning, such as "Data Quest: Big Data Adventure," has proven benefits in motivating students and enhancing their comprehension. A compelling user interface (UI) and user experience (UX) are crucial, as they create an intuitive and immersive environment that captures students' attention and maintains engagement throughout the learning process [28], [29]. For example, a UI that effectively organizes information and a UX that facilitates smooth navigation help students focus on learning without unnecessary distractions, enabling a seamless journey through the educational content.

An engaging game storyline and mechanics also play a significant role in reinforcing students' interest and curiosity [30], [31]. By progressing through different arcs representing core concepts, students actively participate in the learning journey, moving from one milestone to another. This form of storytelling, combined with interactive gameplay, makes complex topics like Big Data more accessible, turning abstract ideas into relatable and manageable tasks that align with students' cognitive abilities and real-world applications.

Moreover, incorporating quizzes as part of the game's flow provides immediate feedback, which is essential for effective learning [32]. These assessments not only reinforce learning but also allow students to self-assess and identify areas that need improvement. The Achievement-Based Unlocking System further adds value by rewarding progress and mastery of each concept before moving on to the next, cultivating a sense of accomplishment and encouraging consistent engagement. This system helps ensure that students not only enjoy the process but also retain critical information, leading to improved learning outcomes.

V. CONCLUSION

Based on the results of this study, it is concluded that the gamified educational RPG "Data Quest: Big Data Adventure," with integrated quiz assessments, effectively addresses the research problem by enhancing both students'

comprehension and engagement in Big Data education. The significant improvement in students' pre-and post-test scores indicates that this gamified approach facilitates a deeper understanding of complex Big Data concepts compared to traditional teaching methods. Additionally, the high levels of student interest and learning outcomes observed, particularly in terms of attention, relevance, and satisfaction, confirm that the game successfully made learning both engaging and meaningful, thereby validating all three hypotheses of the study.

This study highlights the benefits of incorporating games in learning, showing that a well-designed educational game with a robust assessment system can improve both student interests and learning outcomes. The RPG elements not only make complex material more accessible but also encourage the application of theoretical knowledge in practical ways, preparing students for real-world applications. Future development could focus on expanding this model to different educational contexts, exploring scalability, and tailoring the game to accommodate various disciplines.

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